Humanoid Robots in Healthcare: Current Systems, Research Frontiers, and Clinical Opportunities

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As of early 2026, modern commercial humanoid robots—bipedal, dexterous, and large-language-model-capable—have demonstrated clinical task execution in controlled laboratory settings, including teleoperated bimanual procedures across seven clinical tasks [4] and autonomous surgical assistance in a cadaveric sphenoidectomy [16]. Yet no full-size bipedal humanoid robot holds FDA clearance or CE marking for any clinical application, and no registered clinical trial has enrolled patients for a study involving such a platform. This review is the first systematic synthesis of the 2022–2026 humanoid-in-healthcare literature through a clinical deployment lens. We surveyed five databases (arXiv, PubMed, IEEE Xplore, ACM Digital Library, and Google Scholar), identifying 14 qualifying peer-reviewed papers; assessed capabilities for 13 commercial and 12 academic platforms; analyzed regulatory pathways under current FDA, EU MDR, and ISO frameworks; and applied the ISO 16290:2013 Technology Readiness Level (TRL) framework to five clinical application domains. All domains are at TRL 3–4; the evidence base is concentrated on a single commercial platform (Unitree G1) and a single research cluster (UC San Diego Yip Laboratory and Johns Hopkins University). Three near-term roles have the strongest case for advancement to TRL 5 within five years: telemedicine and telepresence clinical assistant, rehabilitation movement demonstrator, and structured eldercare companion and physical assistant. The primary barriers are not platform capability but the absence of accepted safety standards for bipedal gait in patient-proximate environments and the absence of registered human-subject feasibility studies—the two gaps that must be resolved, in that order, before regulatory engagement can begin.

CCS Concepts: • **Computer systems organization** → **Robotics**; • **Computing methodologies** → *Artificial intelligence*; • **Social and professional topics** → *Medical informatics.*

Additional Key Words and Phrases: humanoid robots, healthcare robotics, embodied AI, large language models, human-robot interaction, clinical deployment, medical robots, technology readiness level, regulatory pathway

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AI Artificial Intelligence
LLM Large Language Model
VLM Vision-Language Model
VLA Vision-Language-Action Model
HRI Human-Robot Interaction
TRL Technology Readiness Level
ADL Activities of Daily Living
FDA U.S. Food and Drug Administration
ROS Robot Operating System
ICU Intensive Care Unit
AOT Action Observation Therapy
RCM Remote Center of Motion

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PFL Power and Force Limiting
IRB Institutional Review Board
EHR Electronic Health Record
DOF Degrees of Freedom

1 Introduction

The most capable humanoid robots ever built are commercially available today. Figure 02, Boston Dynamics Atlas (electric), Tesla Optimus Gen 2, Unitree G1, Fourier GR-2, and more than a dozen comparable platforms combine bipedal locomotion, dexterous multi-fingered manipulation, and onboard sensing into systems that operate in spaces designed for humans, handle objects designed for human hands, and respond to instructions in natural language. Twelve of thirteen major commercial humanoid platforms now integrate large language models directly into their control architecture. In the laboratory, a Unitree G1 teleoperated by a non-clinician completed seven distinct clinical procedures—auscultation, ultrasound, IV catheter insertion, and four others—in a controlled experimental setting [4]; a second system operated as a surgical first assistant in a cadaveric bilateral sphenoidectomy [16]. Yet not one of these systems holds FDA clearance or CE marking for any clinical application. No registered Phase I, II, or III clinical trial has enrolled a patient for a study involving a full-size bipedal humanoid platform. The technical capability exists at a level that warrants serious clinical investigation. The clinical deployment does not. This review examines why.

The disparity between demonstrated capability and clinical deployment is not a permanent condition; it is a developmental phase. Two technological developments converged after 2022 to move the humanoid healthcare question from speculative to tractable. The first is hardware: commercial humanoid platforms crossed the payload, dexterity, and locomotion threshold required to attempt clinical and care tasks in their natural environments. Actuator and sensor configurations now exist that allow manipulation of clinical instruments—stethoscopes, ultrasound probes, laparoscopes—and navigation of clinical spaces at human working height and stride. The second is software: the integration of large language models and vision-language-action (VLA) models removed the per-task explicit programming requirement that previously made general-purpose humanoid deployment impractical. Instruction-following, task generalization, and adaptive response to unstructured environments became achievable without the months-long engineering cycles that earlier control architectures demanded. Both were necessary; neither alone was sufficient. The result is a field that moved, in roughly three years, from proof-of-concept locomotion demonstrations to laboratory execution of clinical procedure sequences. The literature reviewed here covers that period: 2022 to early 2026.

The existing review literature does not yet address this convergence. The closest prior work is Cunha et al. [17], a 2025 scoping review of 27 studies on care provided by humanoid robots, with a search cutoff of November 2023—predating the LLM/VLA integration that defines the current period entirely. Earlier reviews of humanoid robots in healthcare [5, 40] and service robots for healthcare delivery [21, 38, 46] treat humanoid and non-humanoid platforms interchangeably, obscuring the form-factor-specific barriers that determine clinical deployability. Embodied AI reviews [32, 33, 37] cover the healthcare vertical as one application among many, without the platform-specific and regulatory specificity that the clinical deployment question requires. This review addresses the gap directly: it is the first synthesis of the 2022–2026 humanoid-specific clinical and care literature that integrates a platform capability assessment, a regulatory pathway analysis grounded in current FDA and EU MDR frameworks, and a per-domain technology readiness evaluation using ISO 16290:2013 TRL criteria—with clinical deployment as the organizing question throughout.

The scope of this review is defined by platform type. Included are bipedal, free-standing humanoid robots of approximately 150–200 cm height, designed for general-purpose manipulation and locomotion in human-scale environments, with literature coverage from 2022 to March 2026. Several categories are excluded with specific rationale. Wheeled social robots—including Pepper (SoftBank Robotics) and ARI (PAL Robotics)—are excluded because their mobility constraints and form-factor dynamics differ from bipedal platforms in ways relevant to clinical deployment analysis; where a wheeled platform appears in a reviewed paper, it is retained with an explicit scope caveat. Sub-100 cm research platforms including NAO are excluded for the same operational envelope reason. Surgical robotic arms (da Vinci) are excluded as task-specific fixed-base systems with entirely separate regulatory histories. Powered exoskeletons are excluded as patient-worn assistive devices operating under a different deployment paradigm. RHP Friends [6], a purpose-built bipedal nursing humanoid, is the only non-commercial academic platform retained without caveat; it meets all inclusion criteria.

This review provides four contributions. First, a systematic survey of 14 qualifying peer-reviewed papers documenting experimental evidence for humanoid robot performance in clinical and care tasks—the most complete such survey to date, covering arXiv, PubMed, IEEE Xplore, ACM Digital Library, and Google Scholar from 2022 through early 2026. Second, a structured capability assessment of 13 major commercial and 12 academic humanoid platforms with a capability comparison table and medical deployment status verified against primary sources; the finding that no platform holds clinical regulatory clearance is itself a data point. Third, a per-domain technology readiness assessment applying the ISO 16290:2013 TRL framework to five clinical application domains—identifying the specific evidence or engineering achievement required to advance each domain. Fourth, a regulatory pathway analysis documenting the current state of FDA, EU MDR, and ISO frameworks as they apply or fail to apply to clinical humanoids, and identifying the jurisdictional gaps that must be resolved before first regulatory clearance can occur.

The paper is organized as follows. Section 2 provides background on the healthcare imperative, the humanoid form-factor rationale, and the LLM/VLA inflection point. Section 3 surveys commercial and academic humanoid platforms. Section 4 reviews the clinical and care research literature by application domain. Section 5 analyzes the technical and translational barriers. Section 6 synthesizes a readiness assessment, identifies near-term deployable roles, specifies research priorities, and outlines the regulatory roadmap. Section 7 concludes.

2 Background

This section establishes the contextual foundations for the survey. We describe the healthcare imperative that motivates increased automation (Section 2.1), explain why the humanoid form factor is particularly relevant to clinical environments (Section 2.2), characterize the inflection point created by large language and vision-language-action models (Section 2.3), trace the historical trajectory of humanoid robotics (Section 2.4), and situate humanoid robots relative to the existing clinical robot landscape (Section 2.5).

2.1 The Healthcare Imperative

The global healthcare workforce is under structural strain that technology alone cannot resolve, but that technology may help mitigate. The World Health Organization projects a shortage of approximately 10 million healthcare workers by 2030, concentrated primarily in low- and middle-income countries but with significant pressure also on high-income systems where demographic aging compounds absolute workforce insufficiency [65]. The root cause is well understood: populations in Organisation for Economic Co-operation and Development (OECD) countries are aging faster than working-age cohorts are growing [45]. In Japan—the most extreme case—persons aged 65 and over already represent approximately 30% of the total population, a proportion that continues to rise under below-replacement fertility rates [11]. The METI 2023 robotics roadmap explicitly identifies nursing-capable robots as a national strategic target by 2030, reflecting governmental acknowledgment that the eldercare workforce gap is structurally unsolvable by recruitment alone.

Nursing workforce shortages present a compounding challenge [29]. The care gap operates at two intersecting levels.

First, the absolute number of care hours demanded rises with the prevalence of chronic disease, multimorbidity, and frailty in an older population: an individual with dementia and comorbid heart failure requires qualitatively more nursing time than a healthy adult, and the number of such individuals is growing rapidly. Second, the tasks that constitute professional care—physical repositioning of patients, vital sign monitoring, medication administration, rehabilitation exercise guidance, and social engagement—are labor-intensive and resist substitution by currently deployed information technology. Electronic health record systems and remote monitoring platforms reduce administrative overhead but do not reduce the need for physical presence at the bedside.

Automation is increasingly discussed as a necessary complement—not a replacement—for human care. The framing matters: the goal is not to displace nurses or physicians, but to extend their effective reach by delegating tasks that are physically demanding, repetitive, or logistically routine, freeing trained clinicians for higher-order judgment and human connection. Wheeled mobile robots already handle medication transport and specimen delivery in hundreds of hospitals. Exoskeletons assist rehabilitation and reduce caregiver musculoskeletal strain. Social robots provide cognitive stimulation and companionship in dementia care settings. Each of these automation layers is real and documented. What remains largely unrealized is a robot capable of performing the physical care tasks—patient repositioning, standing assist, equipment handling, clinical examination—that currently require a human body in proximity to the patient. This is where the humanoid form factor becomes relevant.

2.2 Humanoid Form Factor: Rationale for Healthcare

For the purposes of this survey, a *humanoid robot* is defined as a bipedal, free-standing robot with a human-proportioned morphology: head, torso, two arms, and two legs, typically ranging from 150 to 200 cm in height. This definition explicitly excludes social robots such as Pepper (120 cm, wheeled base) and NAO (58 cm, child-scale), surgical manipulator arms such as the da Vinci system, and wearable exoskeletons. The rationale for this scope restriction is not hierarchical—social robots and surgical robots are clinically important and are discussed in Section 2.5—but conceptual: the form-factor hypothesis is specifically about whether human-scale bipedal morphology confers advantages in human-built environments that smaller or non-mobile systems cannot provide.

Three distinct lines of argument support the hypothesis that the humanoid form factor is particularly promising for healthcare deployment.

Environmental compatibility. Hospitals, care homes, and residences are built for human-sized bipedal operators. Doorways, corridors, stairways, beds, examination tables, supply cabinets, and clinical instruments are all dimensioned for a person approximately 160–180 cm tall with two-handed reach. A bipedal humanoid can, in principle, navigate these environments without infrastructure modification: walk through standard doorways, climb stairs, reach across a standard hospital bed, and operate existing clinical tools (stethoscope, bag-valve mask, otoscope, ultrasound probe) designed for human hands. The technical demonstration by Atar et al. [4] explicitly grounds this observation: the Unitree G1 demonstrated seven medical procedures using existing clinical instruments, without facility modification.

Neuroscientific basis for human-robot interaction. Action observation therapy (AOT)—a rehabilitation technique in which patients observe biological motion to activate sensorimotor circuits—rests on the mirror neuron hypothesis:

observing purposive action activates similar neural circuits as performing that action [51]. Nguyen et al. [41] provide the most direct empirical evidence available: EEG recordings from healthy participants during observation of a humanoid robot performing reaching and grasping actions showed sensorimotor response patterns comparable to those elicited by observing a human agent perform the same actions. This finding provides neurophysiological grounding for the hypothesis that the humanoid form specifically may be required for AOT efficacy. A complementary behavioral finding comes from Sayis and Gunes [54], who demonstrated that only the humanoid robot condition—not a voice-only assistant—produced improvements in mood and increased self-disclosure in a therapeutic writing study (n = 42), attributing the difference to the physical embodiment of the humanoid.

User acceptance and eye-level interaction. Patients in clinical settings are frequently in positions of physical vulnerability. The SPRING project [2] (*scope caveat*: ARI is a wheeled platform, not bipedal) found that users in a real geriatric day-care facility explicitly cited upright, eye-level conversational presence as important for dignified interaction—a form-factor finding directly relevant to bipedal humanoid design regardless of the mobility substrate. Additional contextual evidence comes from research on the HRP-4C platform, which demonstrated that gait naturalness and facial expression generation systematically affected observer ratings of approachability and social presence [14].

These findings collectively support—but do not yet conclusively prove—the form-factor hypothesis at clinical scale. The evidence is encouraging; it is not sufficient to substitute for clinical validation.

2.3 The LLM/VLM/VLA Inflection Point

Prior to approximately 2022, humanoid robots required explicit, hand-crafted programming for every behavioral task. A robot capable of performing a stethoscope placement procedure had to be programmed, step by step, for that specific procedure. Deploying the same robot for a second procedure required a substantially new programming effort. This paradigm fundamentally limited humanoid deployment in clinical environments, which are intrinsically unstructured.

The emergence of large language models (LLMs), vision-language models (VLMs), and vision-language-action (VLA) models has materially changed this constraint. GPT-4 and its successors demonstrated that natural language can serve as a general-purpose task specification interface, capable of interpreting novel instructions and decomposing them into action plans [44]. Physical Intelligence's π_0 and $\pi_{0.5}$ models introduced flow-based VLA architectures mapping language instructions and visual observations directly to robot actions, trained on thousands of hours of real robot demonstrations [7]. OpenVLA [27] demonstrated that open-source, community-accessible VLA models could achieve meaningful generalization from limited task demonstrations. RT-2 [9] established that vision-language foundation models trained on internet-scale data transfer task-relevant knowledge to robotic control without task-specific fine-tuning. Embodied AI surveys covering the healthcare vertical confirm that this software inflection is accelerating clinical relevance [32, 33, 37].

Yuan et al. [66] demonstrated this integration paradigm in a dementia care context, combining reinforcement learning with a large language model to generate adaptive, personalized robot behavior. Although Pepper is a wheeled robot (not within our bipedal scope), the LLM integration framework—specifically its ability to model patient-specific cognitive-emotional states and adapt behavior accordingly—is directly applicable to bipedal humanoid platforms.

The commercial humanoid wave of 2022–2026 coincides with this LLM inflection, and the coincidence is not accidental. Twelve of the thirteen commercial humanoid systems surveyed in Section 3 have confirmed or announced LLM/VLA integration as of 2026. The design assumption of current commercial humanoids is language-conditioned, generalist behavior: these systems are not being built as specialized single-task machines, a prerequisite for clinical task diversity.

2.4 Historical Evolution of Humanoid Robotics

The history of humanoid robotics over the past three decades can be organized into three eras, followed by the nascent healthcare entry period this review covers.

Locomotion proof-of-concept era (1996–2010). Honda's ASIMO (2000) demonstrated that dynamic bipedal walking was achievable with then-current actuation and control technology [14, 53]. The AIST HRP series (HRP-2 through HRP-5P, 1998–2019) established the research infrastructure for whole-body motion planning [26]. Preview control of the zero-moment point (ZMP) provided the mathematical foundation for stable bipedal gait generation [25]. These systems proved that bipedal locomotion was a solvable engineering problem, with no meaningful manipulation capability or path to clinical relevance. ASIMO was retired in 2022.

Manipulation and integration era (2010–2020). The DARPA Robotics Challenge (DRC, 2012–2015) revealed the gap between locomotion capability and robust task execution. Boston Dynamics' hydraulic Atlas became the reference platform for whole-body manipulation research and produced dynamic capability demonstrations while revealing that force-controlled manipulation at clinical precision remained largely unsolved [14]. The decade consolidated learning-based locomotion control, particularly deep reinforcement learning policies trained in simulation and transferred to hardware. Recent work demonstrates that transformer-based policies trained on motion-capture data enable robust zero-shot locomotion on commercial humanoids [49].

Commercial scale-up era (2022–present). The confluence of electric actuator maturation, learning-based locomotion policies, and LLM integration produced the current commercial humanoid wave. Thirteen distinct commercial humanoid systems reached hardware maturity between 2022 and 2026. Global investment in humanoid robotics grew rapidly through 2024, driven by the convergence of electric actuator maturation and LLM integration, with confirmed commercial deployments concentrated in automotive and logistics applications. As of March 2026, none of these thirteen systems has been deployed in a confirmed clinical patient care context.

Healthcare entry point (2024–2026). Atar et al. [4] demonstrated bimanual teleoperation of a Unitree G1 across seven clinical procedures with approximately 70% aggregate success—the first systematic evaluation of a current-generation commercial humanoid against a clinically meaningful task battery. Cho et al. [16] extended this to a cadaveric endoscopic sinus surgery at Johns Hopkins University—the first documented use of a modern commercial bipedal humanoid in an actual (cadaveric) surgical procedure. This survey covers this nascent period.

Figure 1 situates these milestones in the broader thirty-year arc of humanoid robot development.

2.5 Humanoid Robots vs. Non-Humanoid Alternatives in Healthcare

A principled comparison with existing clinical robot categories is necessary to locate the humanoid value proposition precisely.

2.5.1 Social Robots: NAO, Pepper, PARO. Social robots have the deepest clinical evidence base in robotics. NAO (SoftBank/Aldebaran, 58 cm, 25 DOF) has accumulated over 1,000 peer-reviewed publications, with clinical use spanning autism spectrum disorder therapy, elderly care, and hospital patient reception [10]. Pepper has been deployed in approximately 2,000 Japanese healthcare facilities for dementia cognitive stimulation and companion interaction, with embodied conversational agent frameworks increasingly integrated for clinical applications [17, 48, 59]. The PARO therapeutic seal robot has demonstrated psychosocial benefits in randomized controlled trials with elderly residents [64]. These platforms set the deployment high-water mark for robots in clinical care settings [1]. However, their form-factor

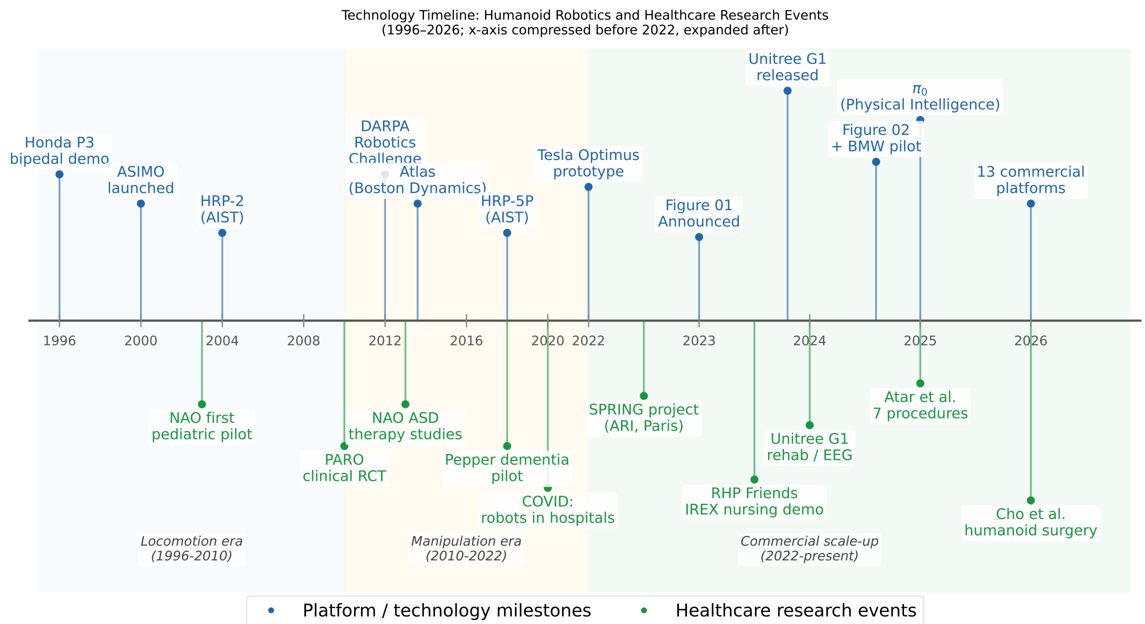

Fig. 1. Technology timeline: humanoid robotics platform milestones (upper track, blue) and healthcare research events (lower track, green), 1996–2026. Shaded regions demarcate three developmental eras: locomotion proof-of-concept (1996–2010), manipulation and integration (2010–2022), and commercial scale-up (2022–present). Healthcare research events lag platform capability by approximately two years in each era.

constraints are real: NAO at 58 cm cannot reach adult-height work surfaces or perform physical care tasks for adults; Pepper on a wheeled base cannot navigate stairways or use standard clinical beds.

2.5.2 Surgical Robots: da Vinci. The da Vinci Surgical System is FDA-cleared (Class II, 510(k)) with over 10 million procedures completed across urology, gynecology, and general surgery [28, 57]. Robotic surgery adoption has grown substantially over the past decade, though adverse event analysis reveals the importance of systematic safety monitoring even for established platforms [3]. The historical trajectory from laboratory prototype to clinical instrument provides context for humanoid deployment [13]. Surgical robots are not competitors to humanoid clinical assistants; they are complementary systems addressing a fundamentally different need [38]. Da Vinci is stationary, surgeon-controlled, and single-purpose. A humanoid clinical assistant operating in a post-surgical ward addresses the nursing and logistics tasks that da Vinci cannot.

2.5.3 Exoskeletons: ReWalk, HAL, Ekso Bionics. Wearable robotic exoskeletons are FDA-cleared (Class II) for spinal cord injury rehabilitation, with Cochrane-level evidence supporting electromechanical-assisted training for walking recovery after stroke [36]. These rehabilitation [59]. Exoskeletons assist the patient's own movement and are not substitutable with a humanoid assistant: they are worn by the patient, require near-ambulatory capacity, and cannot lift or reposition a non-ambulatory patient. The GentleHumanoid framework [34] demonstrates the humanoid's complementary role: learning upper-body compliance for sit-to-stand assistance addresses patients for whom exoskeletons are not an option.

2.5.4 *Wheeled Mobile Robots: TUG, Moxi.*

Autonomous wheeled mobile robots address hospital logistics with confirmed operational deployments at major US health systems [21, 38]. These systems are purpose-built for point-to-point transport on flat pre-mapped surfaces. They cannot navigate stairs, interact with clinical equipment not specifically engineered for their interfaces, or perform physical patient care tasks. The humanoid argument relative to wheeled logistics robots is a division of labor argument: wheeled robots solve the logistics layer effectively; humanoids are needed for the physical assistance and care layer.

2.6 Section Summary

The four alternative categories collectively frame a finding that motivates this review: no existing clinical robot category combines (a) general ambulatory mobility in human-built environments, (b) bimanual dexterous manipulation of existing clinical tools, and (c) natural language-mediated task specification enabling adaptive behavior in unstructured settings. This combination is the humanoid value proposition for healthcare. The evidence reviewed in this section characterizes it as *particularly promising*, not as *clinically proven*. The remainder of this survey characterizes where the field stands, what has been demonstrated, and what must be resolved before this promise can be realized.

3 Current Humanoid Systems

The global humanoid robot landscape in 2024–2026 comprises two largely distinct populations: commercial platforms designed for industrial labor and a set of academic research systems whose provenance spans two decades of bipedal locomotion research. The central empirical finding of this section is that no confirmed clinical deployment of any full-size bipedal humanoid robot exists as of March 2026 [4, 16, 17].

3.1 Commercial Platforms

The commercial humanoid sector underwent a decisive architectural transition between 2022 and 2024. Earlier systems relied on hydraulic actuation for its high force density, at the cost of mechanical complexity, fluid-system maintenance, energy inefficiency, and noise. The generation entering commercial deployment in 2024–2025 is uniformly electric, with brushless DC motors or high-torque servo actuators. This transition has reduced unit cost, simplified field maintenance, and enabled quieter operation—properties that are prerequisites for hospital environments. A second defining characteristic of the 2024–2026 commercial generation is LLM-native design: twelve of the thirteen systems surveyed were designed to support VLA models or LLM interfaces for task specification [14].

Tier 1 — Production-scale deployment.

Figure 02 (Figure AI, USA; August 2024) stands at 168 cm and 70 kg, with 28 total degrees of freedom (DOF) and 16 DOF per hand pair. Payload capacity is 20–25 kg; battery runtime is approximately 5 hours. The documented deployment is a BMW manufacturing facility (Spartanburg, SC), contributing to the assembly of over 30,000 vehicles over 11 months. No healthcare deployment has occurred.

Tesla Optimus Gen 2 (Tesla, Inc., USA; December 2023) measures 173 cm and 57 kg, with 28+ body DOF and 11–22 hand DOF. The platform runs Tesla's Full-Self-Driving-derived neural network stack. Current deployments are internal to Tesla's factories, performing battery cell sorting and parts handling. Publicly available documentation does not confirm any clinical deployment or regulatory submission for medical applications.

Boston Dynamics Atlas (electric) (Boston Dynamics/Hyundai Motor Group; unveiled April 2024) weighs 89 kg with 28 DOF and a sustained payload of 30 kg (50 kg instantaneous). A strategic partnership with Google DeepMind (announced CES January 2026) is integrating Gemini Robotics VLA foundation models. Target deployment is automotive manufacturing. No medical or hospital deployment has occurred.

Agility Digit v4 (Agility Robotics/Amazon, USA; commercial deployment 2024) stands 175 cm and 65 kg, with approximately 3 DOF per arm and 16 kg payload. Confirmed deployments include GXO Logistics (Flowery Branch, GA; over 100,000 totes moved) and Amazon fulfillment center pilots. Healthcare has been cited as a long-term target but no clinical deployment has occurred.

Unitree G1/H1 (Unitree Robotics, China) are the most important commercial systems for this paper's scope. The H1 (2023) stands 180 cm and 47 kg with 19 DOF; the G1 (May 2024) is 132 cm and 35 kg, with 23–43 DOF (base to EDU variant), priced at approximately $13,500–16,000. Unitree is additionally notable as the only major commercial manufacturer to release and maintain a fully open-source SDK. The G1 is the **only commercial humanoid with documented medical procedure research**: Atar et al. [4] at the UC San Diego Yip Lab demonstrated seven medical procedures with approximately 70% task success; Cho et al. [16] employed a Unitree G1 as first assistant in endoscopic surgery. Both constitute academic research prototypes, not clinical deployments. No Unitree robot has been cleared or deployed as a clinical tool in any healthcare system.

Tier 2 — Early commercial.

1X NEO Beta (1X Technologies, Norway; August 2024) is the lightest platform at 30 kg, with 75 total DOF and 22 DOF per hand, noise specified at 22 dB—a potentially relevant characteristic for patient care environments. *Apptronik Apollo* (Apptronik, USA; August 2023) stands 173 cm and 72.6 kg, with 25 kg arm payload; current pilots are in automotive assembly (Mercedes-Benz). *Sanctuary Phoenix Gen 8* (Sanctuary AI, Canada; January 2025) is 170 cm and 70 kg with 21 DOF per hand and 5 mN tactile force sensitivity—the most capable haptic sensing in the commercial tier—driven by the proprietary Carbon AI system developed with Microsoft. No Tier 2 system has a confirmed clinical deployment.

Tier 3 — Development and pre-commercial.

Fourier GR-2 (Fourier Intelligence, China; late 2024) stands 175 cm and 63–65 kg with 53 total DOF and 12 hand DOF per pair, with six tactile sensor arrays. Fourier Intelligence has a healthcare-affiliated subsidiary with confirmed research collaborations at Shirley Ryan AbilityLab and Tan Tock Seng Hospital; however, these partnerships involve exoskeleton devices, not the GR-2 humanoid. *UBTECH Walker X* (UBTECH Robotics, China) is 130–145 cm and 63–77 kg with 41 DOF and multimodal LLM integration; UBTECH maintains a separate healthcare product line distinct from Walker X. *AgiBot A2* (AgiBot/Zhiyuan, China; 2024) is 175 cm and 55 kg with 49+ DOF and an estimated 5,168 units shipped in 2025 (all industrial and service contexts). *NEURA 4NE-1* (NEURA Robotics, Germany; 2023–2024) is 170–180 cm with confirmed force-torque sensing at 0.1 N sensitivity. No Tier 3 system has a confirmed clinical deployment.

Deployment reality. All thirteen commercial systems surveyed are deployed exclusively in industrial or logistics contexts as of March 2026. The sole departure from this pattern is the Unitree G1's use as a research prototype in two academic medical studies [4, 16]. The gap between demonstrated capability in controlled industrial tasks and validated safety in patient-contact clinical environments remains the central barrier to medical deployment.

3.2 Academic and Research Platforms

RHP Friends (CNRS-AIST Joint Robotics Laboratory/Kawasaki Heavy Industries, Japan/France) is the only academic humanoid explicitly designed and demonstrated for nursing care [6]. At 168 cm and 54 kg with 30 DOF, it is human-scale and directly deployable in standard clinical environments. Its defining architectural feature is seamless switching between autonomous and teleoperated control: routine tasks execute autonomously while non-routine or high-stakes tasks hand off to a remote human operator without observable interruption. The system was demonstrated at IREX

Table 1. Humanoid Platform Capability Comparison (March 2026). N/A = not publicly disclosed. †Research prototype only, not clinical deployment. *Not bipedal under this paper's scope definition; included as deployment benchmark.

Robot	Type	Ht	Arm DOF	Payload	LLM Integration	Open SW	Medical Deployment
Figure 02	Commercial	168	N/A (28 total)	20–25 kg	Yes (in-house VLA)	No	None
Tesla Optimus G2	Commercial	173	N/A (28+ total)	20 kg	Yes (FSD-derived)	No	None
BD Atlas (elec.)	Commercial	∼165	N/A (28 total)	30 kg	Yes (Gemini, announced)	No	None
Agility Digit v4	Commercial	175	∼3/arm	16 kg	Announced	Partial	None
Unitree G1	Commercial	132	N/A (23–43)	N/A	Yes (VLA, open)	Yes	Research proto. [4, 16]†
Unitree H1	Commercial	180	N/A (19 total)	N/A	Yes (open)	Yes	None
1X NEO Beta	Commercial	165	22/hand	25 kg	Yes (proprietary)	No	None
Apptronik Apollo	Commercial	173	N/A	25 kg	Yes (Google AI)	No	None
Sanctuary Ph. G8	Commercial	170	21/hand	25 kg	Yes (Carbon/Microsoft)	No	None
Fourier GR-2	Commercial	175	12/arm pair	N/A	Announced	Partial	None
UBTECH Walker X	Commercial	130–145	7/arm	10 kg	Yes (multimodal LLM)	No	None
AgiBot A2	Commercial	175	7/arm	N/A	Yes (proprietary)	No	None
NEURA 4NE-1	Commercial	∼175	N/A	15–20 kg	Yes (cognitive AI)	No	None
RHP Friends	Academic	168	7/arm	N/A	No	Partial	IREX 2023 nursing demo [6]
NAO*	Academic	58	N/A (25 total)	N/A	No	Partial	HRI/ASD trials (social only)
Pepper*	Academic	120	N/A	N/A	No	Partial	2,000+ Japan care facilities

2023 over three days executing patient transfer assistance and circuit breaker operation. No patients were involved. This constitutes the most directly healthcare-relevant academic humanoid performance documented in the literature.

NAO (Aldebaran Robotics/SoftBank Robotics; V6 current) is the most extensively studied humanoid in healthcare research (over 1,000 peer-reviewed publications). At 58 cm and 5.6 kg with 25 DOF, it has been deployed in ASD therapy trials, elderly care facilities, and hospital waiting areas. Over 19,000 units are distributed globally. *Scope caveat*: NAO's form factor is categorically unsuitable for physical adult care tasks. All clinical evidence for NAO pertains to social and cognitive interaction.

Pepper (SoftBank Robotics/Aldebaran) is the most extensively deployed humanoid-form robot in operational healthcare settings: over 2,000 Japanese care facilities use or have used Pepper for dementia cognitive stimulation and companion interaction. Production was paused in June 2021. *Scope caveat*: Pepper is 120 cm and moves on a three-wheeled omnidirectional base—it is not bipedal and does not meet this paper's scope definition.

iCub/iCub3 (Italian Institute of Technology; 2004–present) is 104 cm and approximately 23 kg with 54 DOF and full-body capacitive tactile skin. iCub3 (2022) adds immersive telepresence capability. Over 40 units are distributed to research laboratories worldwide. No confirmed clinical deployment exists. Platforms without confirmed healthcare use include TALOS (PAL Robotics), Valkyrie (NASA JSC), WALK-MAN (IIT), TORO (DLR), LOLA (TU Munich), and the discontinued ASIMO (Honda, retired 2022).

3.3 Capability Comparison

Table 1 compares key specifications for all commercial and selected academic humanoid platforms. All values are drawn directly from verified primary sources; unknown or undisclosed values are marked N/A; approximate values are prefixed with ∼. The Medical Deployment column reflects confirmed deployments only.

3.4 Open-Source Software Ecosystem

A survey of prominent open-source repositories relevant to humanoid robotics yields a clear finding: no dedicated software stack exists for humanoid medical deployment. What does exist is a maturing ecosystem of general-purpose infrastructure.

Foundation models and VLA frameworks. LeRobot [12] (HuggingFace; 22,179 stars; Apache 2.0) provides a standardized, hardware-agnostic pipeline for training imitation learning and VLA policies with direct hardware support for the Unitree G1. Physical Intelligence's openpi provides the π_0 [7] VLA foundation model pre-trained on over 10,000 hours of real robot demonstrations, with state-of-the-art results on ALOHA bimanual manipulation benchmarks [67] and support for fine-tuning with minimal demonstrations. OpenVLA [27] (openvla/openvla; 5,479 stars; MIT), a collaboration of Stanford, Berkeley, and Toyota Research Institute, is a 7B-parameter VLA model trained on 970,000 real-world demonstrations from the Open X-Embodiment dataset [43]. The Diffusion Policy [15] framework provides a complementary visuomotor policy approach based on denoising diffusion models. All four are general-purpose manipulation frameworks with no healthcare-specific training data, safety validation, or clinical task benchmarks.

Simulation environments. Genesis (Genesis-Embodied-AI/Genesis; 28,254 stars; Apache 2.0) is a unified differentiable physics simulation platform released in December 2024. Isaac Lab (isaac-sim/IsaacLab; 6,544 stars; BSD-3-Clause) is NVIDIA's GPU-accelerated framework with direct support for Unitree G1 and H1. Neither platform provides medical environment models, clinical workflow scenarios, or validated patient-contact physics.

Unitree hardware ecosystem. The Unitree GitHub organization maintains nine publicly accessible repositories including unitree_rl_gym (3,022 stars), unitree_sdk2 (932 stars), and xr_teleoperate (1,309 stars) for full-body teleoperation via XR headsets—the software component used in the Atar et al. teleoperation study [4].

The infrastructure gap. Four categories of infrastructure required for clinical deployment are entirely absent from the open-source ecosystem: (1) a validated safety layer enforcing patient-contact force limits at runtime; (2) clinical workflow integration connecting humanoid actions to hospital EHR and nursing systems; (3) an open clinical demonstration dataset (every major VLA training corpus consists of household and tabletop tasks); and (4) a validated medical simulation environment with clinical instrument physics and patient anatomy proxies. The transition from "general-purpose manipulation capability" to "clinically safe, task-verified medical assistance" is primarily a software and validation challenge for which the open-source community has not yet produced foundational components.

Table 2 summarizes the most prominent open-source repositories in this ecosystem.

4 Medical Applications and Research

Despite growing interest in humanoid robots for healthcare, the peer-reviewed evidence base as of early 2026 is remarkably sparse. A systematic search spanning arXiv, PubMed, IEEE Xplore, ACM Digital Library, and Google Scholar, covering 2022–2026, yielded 14 qualifying papers under the bipedal-or-human-form humanoid scope defined in Section 2. The literature is strongly concentrated on a single commercial platform—the Unitree G1—and is dominated by a single research cluster (the UC San Diego Yip Laboratory and Johns Hopkins University), which together account for all three papers in the clinical procedures domain. This concentration is itself a finding: the evidence for humanoid robot capability in clinical settings is real but narrow, and independent replication on alternative platforms has not yet occurred.

4.1 Clinical Procedures and Surgical Assistance

The most substantive cluster of evidence concerns teleoperated humanoid robots performing discrete clinical procedures. Three papers constitute this cluster, all published between 2025 and 2026, all using the Unitree G1.

Bimanual teleoperation for dexterous medical interventions. Atar et al. [4] (UC San Diego Yip Laboratory, PI: Michael Yip) developed a bimanual teleoperation system integrating the Unitree G1 with Inspire Gen4 dexterous hands—four-fingered grippers with 12 DOF each—controlled via bilateral impedance control and real-time motion

Table 2. Open-Source Software Ecosystem for Humanoid and Embodied-AI Robotics (March 2026). Stars are approximate GitHub counts at time of writing. Medical Relevance: **H** = high (direct hardware/algorithm support for clinical candidate platforms), **M** = medium (transferable general capability), **L** = low (infrastructure/simulation only). No repository provides clinical-specific safety validation or medical workflow integration.

Repository	**Org / Author**	**Stars**	**Purpose**	**License**	**Notes**	**Med. Rel.**
lerobot	HuggingFace	~22k	VLA/imitation learning pipeline	Apache 2.0	Supports Unitree G1; no clinical data	H
openpi	Physical Intelligence	~10k	π_0 VLA foundation model	Apache 2.0	10k+ hrs robot demos; bimanual ALOHA	H
openvla	Stanford/Berkeley/TRI	~5.4k	7B-param open VLA model	MIT	970k demos; generalist manipulation	H
act	tonyzhaozh	~5k	Action Chunking Transformers	MIT	Original ALOHA bimanual policy code	H
diffusion_policy	Columbia/MIT	~4.5k	Diffusion-based robot policy	MIT	State-of-art imitation learning	M
Genesis	Genesis-Embodied-AI	~28k	Differentiable physics simulation	Apache 2.0	No medical environment models	M
IsaacLab	isaac-sim (NVIDIA)	~6.5k	GPU-accelerated RL + sim	BSD-3	G1/H1 native support; no clinical env.	M
unitree_rl_gym	Unitree Robotics	~3k	RL locomotion for Unitree robots	BSD-3	G1/H1/B2 locomotion training	H
xr_teleoperate	Unitree Robotics	~1.3k	Full-body XR teleoperation	Apache 2.0	Used in Atar et al. 2025 study	H
unitree_sdk2	Unitree Robotics	~0.9k	Hardware SDK for G1/H1/B2	Apache 2.0	Low-level hardware interface	H
ManiSkill	HKUST	~1.8k	Embodied manipulation benchmark	Apache 2.0	No clinical task benchmarks	M
RoboManip	isri-robot	~0.3k	Bimanual manipulation dataset	MIT	Tabletop tasks only	M
Open Duck Mini	apirrone	~2k	Miniature bipedal robot (open HW)	MIT	Educational; not clinical scale	L

retargeting. The system was evaluated across seven distinct clinical procedures: (1) cardiac and pulmonary auscultation via stethoscope placement, (2) respiratory assistance via bag-valve mask ventilation, (3) Leopold maneuvers for fetal position assessment, (4) otoscopic examination, (5) intravenous catheter insertion, (6) point-of-care ultrasound (POCUS), and (7) blood glucose finger-stick testing. Non-clinician operators achieved an aggregate procedural success rate of approximately 70% in controlled laboratory conditions; no human subjects were involved. Auscultation and ultrasound imaging yielded the strongest performance; IV catheter insertion and finger-stick testing were the most challenging, with failures attributable to force precision requirements beyond the system's current capability.

The paper documents two technical barriers: insufficient force output in the actuator configuration, and sensor sensitivity limitations in the tactile feedback loop. The study is assessed at TRL 4—technology validated in laboratory conditions using representative test fixtures, but not yet in a clinical environment or with patient contact.

Laparoscopic surgery via humanoid teleoperation. Liang et al. [30] (UCSD Yip Laboratory) extend humanoid clinical teleoperation to laparoscopic surgery. Their LapSurgie framework introduces an inverse-mapping strategy adapting the Unitree G1's arm kinematics to drive manual-wristed laparoscopic instruments while enforcing a remote-center-of-motion (RCM) constraint at the trocar site—a non-negotiable geometric safety requirement ensuring the instrument pivots about the entry point rather than exerting lateral forces on the abdominal wall. The authors advance an access-equity argument: the humanoid form factor maps naturally to laparoscopic ergonomic conventions, potentially enabling specialist surgical capability in geographically underserved settings.

Humanoid robot as surgical first assistant. Cho et al. [16] (Johns Hopkins University) report the most clinically significant result in this review: a teleoperated Unitree G1 serving as first assistant in a bilateral cadaveric sphenoidectomy performed by an attending otolaryngologist. The robot held and repositioned an endoscope under the surgeon's direction, providing the continuous endoscopic visualization that a human first assistant would normally supply. This is the first documented use of a modern commercial bipedal humanoid in a real surgical procedural context. Cadaver studies are a recognized pre-clinical methodology representing the step immediately preceding IRB protocols with human subjects.

The paper identifies specific engineering targets for clinical translation: endoscope stability, compliant force control during anatomical contact, and autonomous scope positioning.

Platform concentration as a structural limitation. All three papers in this subsection use the Unitree G1; P1 and P2 originate from a single laboratory under a single principal investigator. Independent replication on other platforms has not yet occurred, limiting generalizability of the current evidence base.

4.2 Elderly and Nursing Care

Purpose-built nursing humanoid. Benallegue et al. [6] document RHP Friends, the only bipedal humanoid platform in the literature designed with nursing assistance as its explicit primary application. Its defining architectural contribution is seamless transparent switching between autonomous and teleoperated control: routine tasks execute autonomously while non-routine tasks hand off to a remote human operator without observable interruption. The system was demonstrated at IREX 2023 performing patient transfer assistance and circuit breaker operation. No patients were involved. TRL 4 for demonstrated hardware capability in a structured exhibition setting.

Large-scale deployment evidence from a wheeled platform. *Scope caveat*: the SPRING project [2] used the ARI robot by PAL Robotics, which operates on a wheeled differential-drive base; it does not qualify as a bipedal humanoid and must not be cited as evidence for bipedal humanoid deployment. Over a four-year EU H2020 program (2020–2024), ARI was deployed in a Parisian geriatric day-care facility with more than 60 older adult participants. Acceptability was positive; participants specifically cited upright posture and eye-level conversational positioning as important for dignified interaction. The SPRING project represents the deployment robustness and user acceptance threshold that fully bipedal systems must replicate.

Institutional stakeholder perspectives. Imtiaz and Khan [22] surveyed 269 nursing home and long-term care administrators across the United States on humanoid robot deployment. Administrators acknowledged potential value in resident engagement and staff support. Primary barriers cited were cost, perceived risk of reduced human contact, and the absence of demonstrated clinical effectiveness data—objections that are economic, ethical, and evidence-based rather than technical.

User-centered design methodology. Ghosh et al. [18] report a user-centered design study combining a NAO humanoid with non-immersive virtual reality for non-pharmacological apathy management (14 participants). *Scope caveat*: NAO is 58 cm and not a full-size bipedal humanoid. The study's contribution is methodological: iterative co-design with care staff improved system usability across design cycles, demonstrating that clinical staff can be engaged as design partners.

4.3 Rehabilitation

Neuroscientific basis for humanoid-specific rehabilitation. Nguyen, Anand, and Johnson [41] report a pilot EEG study measuring sensorimotor neural responses in healthy participants during observation of a humanoid robot versus a human actor performing reaching and grasping actions. The central finding is that common neural response patterns consistent with mirror neuron system activation are elicited by humanoid robot observation at levels comparable to human observation. This is a foundational result for the use of humanoid robots as movement demonstrators in AOT programs for stroke patients. Caveats: pilot study in healthy subjects; clinical replication in patients with motor impairment is necessary. TRL: 3 (experimental proof of concept).

Remote neuromotor rehabilitation via humanoid telepresence. Sobrepera et al. [58] describe a Social Robot Augmented Telepresence (SRAT) system combining a humanoid robot with mobile telepresence for remote neuromotor rehabilitation. A case series of six participants (three adult stroke survivors, three pediatric participants) found that five of six rated the SRAT system superior to standard video telepresence, attributing the preference to the physical embodiment of the humanoid—the ability to demonstrate movement and convey nonverbal cues through body posture. TRL: 4, demonstrated in a small clinical case series.

Compliance for physical assistance tasks. Lu et al. [34] report the GentleHumanoid framework, a spring-based unified compliance architecture for the Unitree G1 enabling safe physical contact during sit-to-stand assistance and upper-body interactions. No patient subjects were involved. The framework's quantified reduction in peak contact forces relative to rigid control constitutes a concrete step toward the force limits specified by ISO/TS 15066 for physical human-robot contact.

4.4 Mental Health and Social Interaction

Autonomous humanoid delivery of wellbeing intervention. Robinson et al. [52] report a pilot randomized controlled trial (n = 230) in which an autonomous humanoid social robot delivered a brief mindful breathing training session to a subclinical general population. Recruitment uptake of 53% indicates the intervention was not aversive. Participants rated the experience as enjoyable and moderately useful. *Caveat*: subclinical general population; robot model undisclosed; primary measure was self-reported experience. TRL: 4 for wellbeing applications in non-clinical settings.

Embodiment advantage in mental health interactions. Sayis and Gunes [54] report a comparative study (n = 42 university students) of humanoid robot versus voice assistant for therapeutic journal writing. Only the humanoid robot condition produced statistically significant mood improvements, increased self-disclosure, and positive perception change. *Caveat*: university wellness population; robot model unspecified. The embodiment advantage finding merits replication in clinical populations.

LLM integration for personalized dementia care. Yuan et al. [66] propose a framework combining a humanoid robot, reinforcement learning, and large language models for adaptive personalized dementia care, introducing probabilistic cognitive-emotional state modeling. *Scope caveat*: this paper uses Pepper, a wheeled robot. It is cited here exclusively for the LLM integration paradigm—cognitive-emotional state modeling adaptable to bipedal platforms. The use of embodied conversational agents for clinical psychology applications is a growing evidence base [48].

Socially assistive robots with automated planning have also been explored in paediatric clinical settings [31], providing methodological precedent for structured social robot interaction protocols.

4.5 Hospital Logistics

No qualifying papers were found for humanoid-specific hospital logistics tasks performed by bipedal humanoid robots. This absence is plausibly explained by two structural factors. First, wheeled autonomous mobile robots already address hospital logistics with proven operational reliability, providing no demonstrated functional advantage for the humanoid form factor in this task category. Second, humanoid hardware development in 2022–2026 was concentrated on high-value industrial contexts, with academic research attention directed toward the more scientifically novel clinical procedure and HRI domains. TRL for bipedal humanoid hospital logistics: 2 (technology concept formulated; no demonstrated examples in relevant contexts).

4.6 Section Summary

Table 3 summarizes the evidence base across the five clinical domains reviewed above.

Table 3. Evidence Summary by Clinical Domain (as of March 2026). TRL per ISO 16290:2013: 1 = basic principles; 2 = concept formulated; 3 = experimental PoC; 4 = lab validated; 5 = relevant-environment validated; 7 = operational prototype; 9 = proven. *ARI/SPRING: wheeled platform, not bipedal; scope caveat applies.

Domain	Key Papers	Platform	Study Type	N	TRL
Clinical procedures	P1 [4], P2 [30], P3 [16]	Unitree G1	Lab; user study; cadaver	0 patients	3–4
Elderly / nursing	P8 [6], P7* [2]	RHP Friends; ARI*	Exhibition demo; field deploy.	0 (demo); 60+*	4; 4–5*
Rehabilitation	P5 [41], P6 [58], P13 [34]	Humanoid; SRAT; Unitree G1	EEG pilot; case series; engineering	0 / 6 / 0	3–4
Mental health / HRI	P11 [52], P14 [54]	Unspecified humanoid	RCT; comparative	230; 42	4–5
Hospital logistics	*No qualifying papers identified; wheeled autonomous systems dominate this role (TRL 2)*				2

The aggregate picture across all five domains is consistent: the field is at TRL 3–4 for bipedal humanoid clinical tasks. Feasibility has been demonstrated in controlled laboratory and pre-clinical settings, primarily by two research groups on a single platform, but no clinical trial with patient subjects, no regulatory submission, and no institutional deployment exists for any full-size bipedal humanoid robot as of early 2026.

Figure 2 visualizes the distribution of the 14 qualifying papers by domain and year, with bubble size proportional to study design strength.

5 Technical and Translational Challenges

The TRL 3–4 state documented in Section 4 reflects a specific constellation of barriers that are neither uniform nor independently solvable. Some are technical: manipulation precision, contact safety during bipedal locomotion, software validation against clinical performance standards. Some are institutional: regulatory pathway ambiguity, liability framework gaps, actuarial absence. And some are structurally circular: regulatory approval requires clinical evidence; clinical evidence requires IRB-sanctioned trials; IRB approval requires safety assurance; safety assurance requires performance standards that regulatory guidance has not yet established. This section diagnoses each barrier specifically.

5.1 Manipulation and Dexterity for Clinical Tools

Atar et al. [4] is the only study to systematically document manipulation performance gaps in a clinical task context. Atar et al. evaluated a Unitree G1 with Inspire Gen4 dexterous hands across seven clinical procedures, achieving approximately 70% aggregate procedural success with non-clinician teleoperators. The paper identifies two specific failure modes: force output insufficiency (the actuator configuration cannot reliably generate the controlled forces required for needle penetration or sustained grip against resistance), and sensor sensitivity limitations (the tactile feedback loop is insufficient for sub-Newton force resolution, preventing detection of the haptic signatures that guide needle procedures in clinical practice).

These findings generalize across the procedure literature. Liang et al. [30] demonstrate that laparoscopic surgery requires enforcing a remote-center-of-motion (RCM) constraint at the trocar site—an engineering workaround for a capability that commercial humanoid arms do not provide natively. Cho et al. [16] identify three engineering targets for endoscopic clinical translation: endoscope stability in confined anatomical cavities, force compliance during tissue contact, and positioning accuracy for reliable camera view.

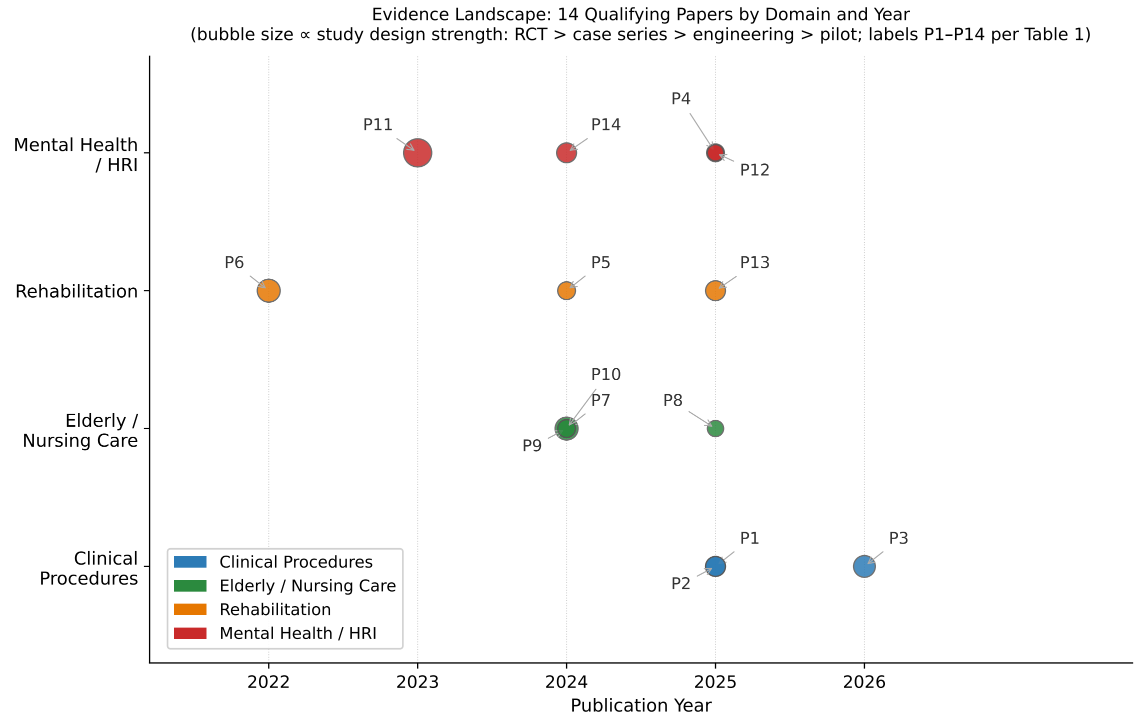

Fig. 2. Evidence landscape: 14 qualifying papers by clinical domain and publication year (bubble size proportional to study design strength: RCT > case series > engineering > pilot). Clinical procedure evidence (blue) is concentrated in 2025–2026 and from a single platform and research cluster. Rehabilitation (orange) and mental health (red) evidence is more temporally distributed but also methodologically limited. No papers qualify for hospital logistics.

Synthesizing across P1, P2, and P3, clinical manipulation requires three capabilities that current commercial humanoid arms do not yet reliably provide: (1) force precision at sub-Newton resolution for needle and catheter procedures [4]; (2) tool-specific kinematic adaptation for laparoscopes, rigid endoscopes, and ultrasound probes; and (3) fingertip-level tactile sensing for grip force discrimination, instrument slip detection, and tissue compliance sensing. Operationally, progress requires force resolution at sub-Newton levels for needle insertion procedures [4], fingertip sensor density exceeding 1,000 sensing elements, and teleoperation round-trip latency at levels consistent with real-time force feedback requirements. Surgical teleoperation research establishes that network latency above 200 ms degrades task completion time and trajectory accuracy in dexterous procedures [35], providing the clinical benchmark for humanoid teleoperation system design. Foundational work on dexterous manipulation documents the multi-dimensional sensor requirements for robust tool handling [42].

5.2 Safe Physical Contact and Bipedal Gait Dynamics

Progress in contact safety has been demonstrated. Lu et al. [34] introduce a spring-based compliance framework for the Unitree G1 that quantifiably reduces peak contact forces during quasi-static tasks including sit-to-stand assistance and upper-body physical contact. This approach builds on the series elastic actuator paradigm [47], in which mechanical spring compliance decouples rapid force transients from the actuated structure. This is the most technically grounded progress toward patient-contact safety in the current literature. Hierarchical whole-body control architectures [55] provide the control-theoretic foundation for multi-priority task execution that safety-critical clinical interaction will require.

The design target for patient-contact force limits is the Power and Force Limiting (PFL) regime of ISO/TS 15066:2016 [24], which provides biomechanically derived thresholds for different body regions: for the face and skull, quasi-static contact force is limited to 65 N and pressure to 110 N/cm^2; for the trunk and thorax, 140 N and 210 N/cm^2. The GentleHumanoid framework demonstrates a Unitree G1 approaching these thresholds for quasi-static upper-body contact.

The critical gap, however, is structural. ISO/TS 15066 was designed for fixed-base collaborative robots—cobots whose base is anchored, whose motion is constrained to a defined workspace, and whose approach velocities are programmed. It does not model contact forces arising from bipedal locomotion. A bipedal humanoid navigating a clinical environment introduces four risk scenarios that ISO/TS 15066 does not bound: (1) balance corrections, in which whole-body torque adjustments generate rapid impulse forces that can contact nearby persons; (2) stumble recovery, in which momentum transfer creates swinging limbs and torso rotation at velocities above the standard's contact speed assumptions; (3) narrow-corridor navigation, in which swinging arms during gait will contact bed rails, IV poles, and patient limbs in spaces with less than one body-width of clearance; and (4) postural transitions, in which a full-size humanoid (35 kg or more) rising to or lowering from a working height near a supine patient generates dynamics that quasi-static force limits do not characterize.

None of the 14 papers reviewed addresses this category of risk, and neither ISO/TS 15066, ISO 13482, nor ISO 10218-1/2 was designed to model or bound gait-induced contact events. This is an open safety engineering problem. Resolution requires either new standards specific to bipedal humanoid robots in patient-contact environments, or architectural solutions—proximity-triggered gait interrupts, patient-detection zones constraining locomotion near occupied beds, whole-body collision prediction for gait trajectories—that have not yet been developed in the published literature.

5.3 Human-Robot Trust and Institutional Adoption

Trust and adoption operate at three distinct layers. The first layer is patient and clinician acceptance. Technology acceptance frameworks established for information systems [63] and their extensions for assistive social robots in elderly care [8, 20] provide the theoretical basis for understanding humanoid adoption barriers. Trust in human-robot interaction is a multidimensional construct influenced by robot performance, human factors, and environmental context [19]. The healthcare-specific question of whether humanoid form factors in clinical settings are perceived as appropriate has been characterized as a strategic and ethical question [46]. Robinson et al. [52] show 53% recruitment uptake for a humanoid wellbeing intervention in a general population (pilot RCT, n = 230). Sayis and Gunes [54] show the humanoid condition was the only condition producing measurable mood improvements in a therapeutic writing task (n = 42). Both results are bounded: the clinical populations central to humanoid deployment differ substantially from general and educational populations in these studies.

The second layer is co-design and workflow integration. Ghosh et al. [18] demonstrate that iterative co-design with care staff improves system usability—a methodological contribution that does not substitute for clinical evidence of safety and effectiveness.

The third and most institutionally consequential layer is the liability gap. No current healthcare liability framework—US tort law, UK clinical negligence, EU product liability—assigns accountability for harm caused by an autonomous robotic action in a care setting. This legal ambiguity deters adoption regardless of technical capability: hospital risk departments cannot quantify institutional exposure for systems operating in legally uncharted territory. Imtiaz and

Khan [22] document this precisely—269 nursing home administrators cite cost, reduction in human contact, and unproven clinical effectiveness as primary barriers.

5.4 Regulatory Pathway

The regulatory challenge is not hostility to this technology class; it is a jurisdictional gap that existing frameworks leave precisely where clinical humanoids sit.

ISO 13482:2014 [23] (Safety Requirements for Personal Care Robots) is the most directly applicable safety standard—it governs robots providing physical assistance in human environments. However, §1.4 of ISO 13482 explicitly excludes devices regulated as medical devices. A humanoid robot performing clinical tasks almost certainly meets the medical device definition under FDA 21 CFR §201(h) and EU MDR Article 2(1). This creates a hard gap: the system is likely too clinical for ISO 13482 but no ISO medical robotics standard covers general-purpose bipedal humanoid assistants.

In the United States, the De Novo classification pathway is the only viable regulatory entry route. No 510(k) predicate exists. A De Novo submission requires the applicant to propose Special Controls from scratch: accepted performance standards for fall prevention in patient-care environments, patient contact force limits for dynamic gait scenarios, teleoperation latency bounds for specific procedure types, and AI/ML software validation methodology consistent with FDA's AI/ML SaMD action plan [61] and December 2024 PCCP guidance [62]. None of these currently exist as accepted standards. This is not procedurally impossible—De Novo has established new device categories for surgical planning software and exoskeletal rehabilitation devices—but it represents a development and regulatory investment no humanoid manufacturer has yet made.

The structural problem is a circular dependency: approval requires clinical evidence; human-subject evidence requires IRB approval; IRB approval requires safety assurance; safety assurance requires performance standards; performance standards require regulatory guidance. The productive path through this circle is a staged evidence generation sequence—in vitro feasibility, cadaver study, IRB-approved human-subject feasibility trial—producing the data needed for a De Novo submission. Cho et al. [16] is the current frontier of that sequence.

In the EU, a humanoid clinical assistant would likely be classified as Class IIa under EU MDR Annex VIII, with autonomous surgical assistance escalating to Class IIb or III. EU AI Act Regulation 2024/1689 additionally classifies any AI-driven clinical humanoid as a high-risk AI system. High-risk AI rules apply from August 2026; AI embedded in regulated medical products transitions to August 2027. Any EU clinical humanoid market entry must satisfy both MDR and AI Act simultaneously—a dual compliance burden for which no completed template yet exists.

Table 4 compares the four principal frameworks across the key dimensions that determine market entry strategy.

5.5 Software Infrastructure and AI Validation

The leading VLA models—openpi/π_0 (10,551 stars), OpenVLA (5,479 stars), and LeRobot (22,179 stars)—represent the highest-capability general-purpose robot learning infrastructure publicly available. For general-purpose humanoid manipulation research, this infrastructure is genuinely strong.

What is absent is the layer between this general infrastructure and clinical deployment. As identified in Section 5.5, the four critical gaps with no current open-source solution are a clinical safety layer, clinical workflow integration, an open clinical demonstration dataset, and a validated medical simulation environment.

The AI validation gap extends to deployment readiness. VLA models achieve high success rates on standard benchmarks, but these share no tasks with the clinical procedure requirements documented in [4]. The distribution shift involves not only visual and geometric differences but also contact physics, tool compliance, and procedural sequencing

Table 4. Regulatory Framework Comparison for Clinical Humanoid Robots (March 2026). PMS = post-market surveillance; PMCF = post-market clinical follow-up; SaMD = Software as a Medical Device; NB = Notified Body. No framework provides humanoid-specific guidance; all entries represent the authors' interpretation of existing rules applied to this novel device class.

Framework	Jurisdiction	Classification	Pathway	Timeline (est.)	Key Requirements / Gap
FDA 510(k)	US	Class I/II	Substantial equivalence to predicate	3–12 months	No predicate exists for clinical humanoid; inapplicable
FDA De Novo	US	Class II (new category)	Novel device; propose Special Controls	12–24 months	Must define fall-prevention, PFL, latency, AI/ML Special Controls from scratch
FDA PMA	US	Class III	Clinical trial evidence required	3–7 years	Overkill for initial classification; may apply for autonomous surgical
EU MDR Class IIa	EU	Medium risk	Notified Body QMS + clinical data	18–36 months	Requires clinical investigation under Art. 62; PMCF post-market
EU MDR Class IIb	EU	Higher risk	Notified Body + full clinical eval.	24–48 months	Autonomous surgical or patient-contact without operator oversight
EU AI Act (High Risk)	EU	High-risk AI	Conformity assessment; Art. 6	From Aug 2026	Parallel obligation to MDR; dual compliance burden for clinical humanoid
ISO 13482:2014	Global	Personal care robot safety	Voluntary standard	N/A	**Explicitly excludes medical devices**; no bipedal gait dynamic PFL spec
ISO/TS 15066:2016	Global	Collaborative robot HRC	Voluntary; used in EU MDR	N/A	Designed for fixed-base collaborative arms; no dynamic gait scenario coverage

requirements categorically distinct from household manipulation. AI in health and medicine more broadly faces similar validation challenges as these systems move toward clinical deployment [50, 60]. Additionally, no humanoid software stack has addressed IEC 62443 Security Level 2 requirements—the minimum threshold for networked systems in hospital environments.

Table 5 synthesizes the five principal technical and translational barriers identified in this section.

6 Discussion and Future Outlook

The evidence surveyed in Sections 4 and 5 presents a field at an inflection point: capable enough to demonstrate clinical relevance in controlled settings, yet short of every threshold—safety engineering, clinical evidence, and regulatory precedent—required for operational deployment.

6.1 Technology Readiness Assessment

The TRL assignments in Table 3 provide the starting point. The aggregate finding—TRL 3–4 across all domains for bipedal humanoid platforms—is not a uniform statement; the TRL ceiling for each domain is governed by a specific domain-particular barrier.

Clinical procedures (current TRL 3–4). Atar et al. [4], Liang et al. [30], and Cho et al. [16] constitute the most technically substantive evidence in the review. The cadaveric study of Cho et al. is the current frontier of the evidence sequence: the step immediately preceding an IRB-approved human-subject feasibility trial. Advancing to TRL 5 requires two things simultaneously: a structured IRB-approved feasibility study in a specific narrow indication (telemedicine-assisted physical examination is the most defensible starting point, because fully teleoperated operation keeps a human in the decision loop at all times), and independent replication on a platform other than the Unitree G1.

Table 5. Technical and Translational Gap Summary (March 2026). Each row identifies a barrier from Section 5, its current state, the threshold required for TRL 5 advancement, which clinical domains it blocks, and the primary evidence or standard defining it. No single gap is insurmountable; they are sequentially dependent in the order listed.

Capability Gap	Current State	Required for TRL 5	Blocking Domains	Key Evidence / Standard
Force precision & tactile sensing	Insufficient sub-N resolution; no fingertip tactile array on commercial platforms	Sub-Newton force feedback; continuous fingertip sensing at clinical manipulation precision	Clinical procedures, Rehabilitation	[4]; [35]
Gait contact safety	ISO/TS 15066 quasi-static thresholds approached for upper body only [34]	Instrumented characterization of gait-induced force profiles; accepted standard for bipedal PFL	All physical-contact domains	ISO/TS 15066 [24]; ISO 13482 [23]
Tool-specific kinematics	No native RCM; no laparoscope-specific axis on commercial arms	RCM enforcement or software analog; tool-path control within trocar constraints	Clinical procedures	[30]; [16]
VLA clinical generalization	VLA benchmarks: household manipulation only; no clinical demonstration dataset exists	Open clinical demonstration dataset; VLA fine-tuned on clinical task protocols with validated performance	Clinical procedures, all autonomous roles	[43]; [67]
Clinical safety standard & regulatory predicate	No accepted standard for bipedal gait in patient-contact environments; no De Novo predicate	ISO scope revision; IRB feasibility study; FDA De Novo with proposed Special Controls	All domains	[23]; [61]; [62]

Elderly and nursing care (TRL 4–5 for social robots; TRL 3–4 for bipedal humanoids). The gap between platform classes is analytically significant. The SPRING project achieved TRL 4–5 through real-world deployment with 60+ older adult users over four years [2]. RHP Friends—the only purpose-built bipedal nursing humanoid—demonstrated patient transfer tasks at IREX 2023 but has not been evaluated in an actual care facility [6]. The path to TRL 5 for bipedal humanoids is not a question of platform capability but of deployment context: a structured pilot in an actual care facility, using SPRING-derived methodology on a bipedal platform, is the specific experiment that would advance this domain.

Rehabilitation (TRL 3–4). Nguyen et al. [41] establish the neuroscientific mechanism; Sobrepera et al. [58] establish clinical acceptability in a small case series; Lu et al. [34] establish the contact-safety engineering foundation. These three papers, from three independent groups, collectively build an evidence case no single paper makes alone. The path to TRL 5 is a powered IRB-approved clinical trial in stroke rehabilitation.

Mental health and HRI (TRL 4–5 in non-clinical populations; TRL 3 for clinical populations). Robinson et al. [52] and Sayis and Gunes [54] achieve methodologically strong results that carry a critical transferability caveat: neither study involves a diagnosed patient population in an active care context. Advancing to TRL 5 requires a feasibility study with a specific patient population where a humanoid delivers a defined therapeutic interaction protocol and outcomes are assessed on validated clinical scales.

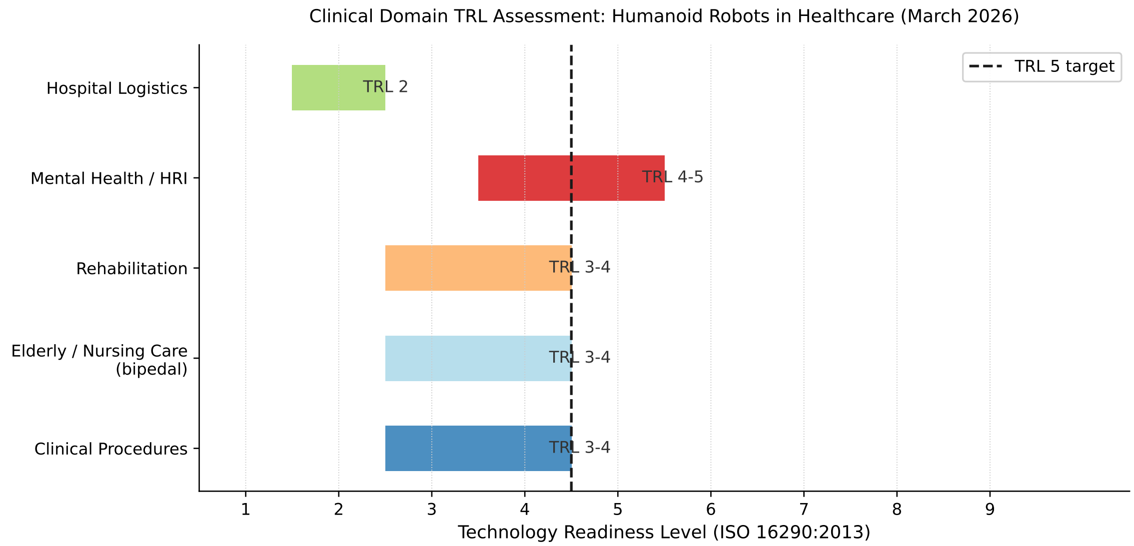

Fig. 3. Technology Readiness Level (TRL) assessment by clinical domain (March 2026). Current TRL range (filled bars) and TRL 5 target (dashed line) per ISO 16290:2013. No domain has reached TRL 7 (operational environment prototype); TRL 5 (relevant-environment validation) is the realistic five-year target for clinical procedures, elderly care (bipedal), and rehabilitation.

Hospital logistics (TRL 2). No peer-reviewed evidence exists for humanoid-specific hospital logistics tasks. Wheeled autonomous mobile robots already address this domain with demonstrated operational reliability. Hospital logistics is not a priority for bipedal humanoid development at this stage.

Cross-domain finding. No domain has reached TRL 7 (system prototype demonstrated in operational environment). TRL 5—the threshold of first human-subject validation in a controlled clinical setting—is the realistic five-year target for the three leading domains.

Figure 3 visualizes the current TRL state and TRL 5 target across all five clinical domains.

Figure 4 quantifies the specific capability gaps between the current state-of-the-art and the required thresholds for two leading clinical deployment domains.

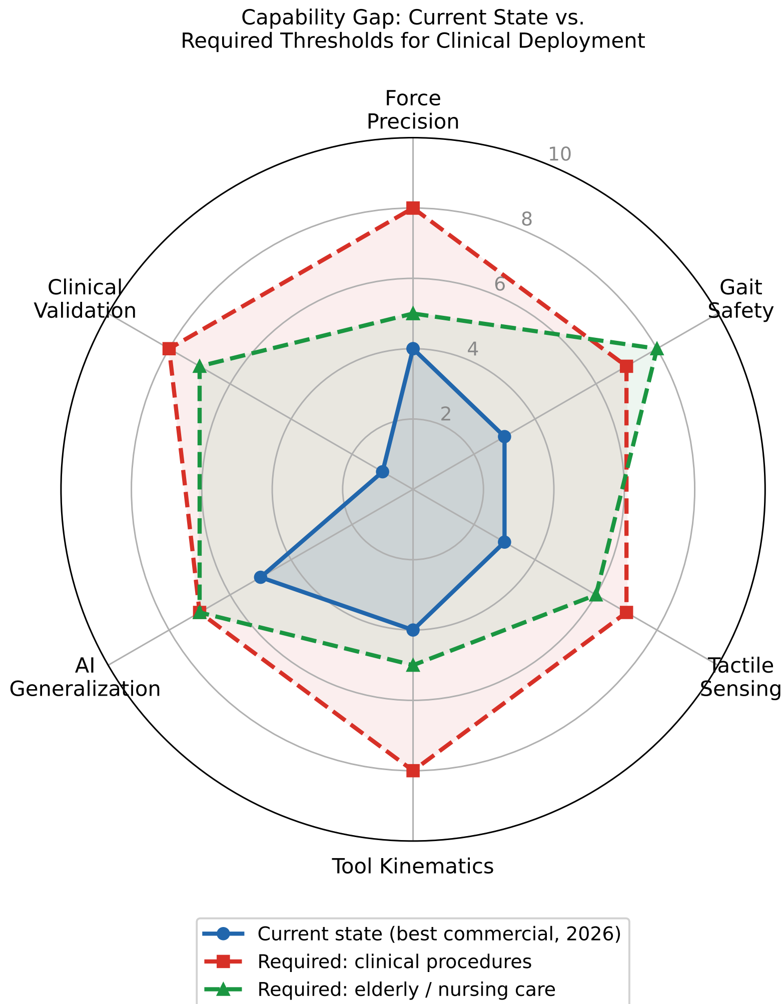

Fig. 4. Capability gap radar chart: current state-of-the-art (blue, 2026) versus required thresholds for clinical procedures (red dashed) and elderly nursing care (green dashed). Scores are on a 0–10 scale representing engineering and evidence maturity. Clinical Validation is the universal bottleneck (current score 1 of 10); Gait Safety is the dominant engineering gap for eldercare; Force Precision and Tool Kinematics are the key gaps for clinical procedure performance.

6.2 Near-Term Deployable Roles

Three roles have the strongest case for TRL 5 achievement within five years.

Role 1: Telemedicine and telepresence clinical assistant. The teleoperated mode is the key risk-mitigation design choice: by keeping a human operator in the decision loop at all times, it avoids the regulatory classification and safety validation burdens associated with autonomous patient contact. The technical infrastructure exists: Atar et al. [4] demonstrate teleoperated clinical task execution across seven procedures, and Liang et al. [30] extend this to laparoscopic surgery. The primary application is specialist access extension—enabling a remote specialist to conduct a physical examination or assist in a procedure in a rural or low-resource setting. Realizing this role at TRL 5 requires a defined physical examination protocol, remote operator training standards, and IRB oversight.

Role 2: Rehabilitation movement demonstrator. This role requires no direct patient contact: the humanoid demonstrates movements that the patient observes or follows. The bipedal gait safety gap identified in Section 5.2—the most universally blocking technical barrier—becomes irrelevant to this specific role, because the safety concern is dynamic contact force during locomotion near patients, not movement observation at distance. The neuroscientific rationale is grounded in P5's mirror neuron evidence [41]; the clinical acceptability is established by P6's case series [58]. Advancing requires a registered feasibility trial adapting a NAO or Pepper protocol to a full-size bipedal humanoid demonstrator in stroke rehabilitation.

Role 3: Structured eldercare companion and physical assistant. RHP Friends [6] is the only platform designed specifically for this role. The hybrid autonomous/teleoperated architecture—in which routine tasks run autonomously and non-routine tasks are handed to a remote operator—is the correct design for this deployment scenario. The SPRING project [2] provides the deployment methodology and user-acceptance evaluation framework. Realizing this role at TRL 5 requires an actual care facility pilot.

6.3 Research Priorities

Four priorities, ordered by their blocking impact on the entire field:

Priority 1: Clinical safety validation for bipedal gait in patient-contact environments. The four gait-induced contact risk scenarios documented in Section 5.2 are not bounded by any existing standard. No IRB will approve human-subject studies involving physical patient contact by a full-size bipedal humanoid without instrumented biomechanical data on these force profiles and an accepted safety standard or architectural safeguard. This is the single most leverage-intensive investment the field can make: resolving it unblocks physical patient-contact studies in all three leading domains simultaneously.

Priority 2: Human-subject feasibility studies in leading applications. The cadaveric endoscopy study of Cho et al. [16] is the current clinical frontier. The specific next step is an IRB-approved, registered feasibility study in a narrow indication with an explicit safety monitor protocol. UC San Diego (Yip Laboratory) and Johns Hopkins are the natural sites.

Priority 3: Open clinical demonstration dataset for VLA training. No public dataset of humanoid robot demonstrations in clinical or clinical-simulated environments exists. Creating an open clinical demonstration dataset—even using simulators and clinical task protocols with mannequins—would unlock foundation model fine-tuning for clinical applications and enable the broader research community to advance clinical policy learning.

Priority 4: Platform diversification for evidence generalizability. The concentration of clinical procedure evidence on the Unitree G1 (P1, P2, P3) is a scientific limitation. Replication of clinical task evaluations on RHP Friends, TALOS, or other platforms would establish whether current findings are platform-specific or generalizable to the bipedal humanoid class.

6.4 Regulatory Roadmap

Figure 5 illustrates the parallel regulatory tracks and their key decision points for both jurisdictions.

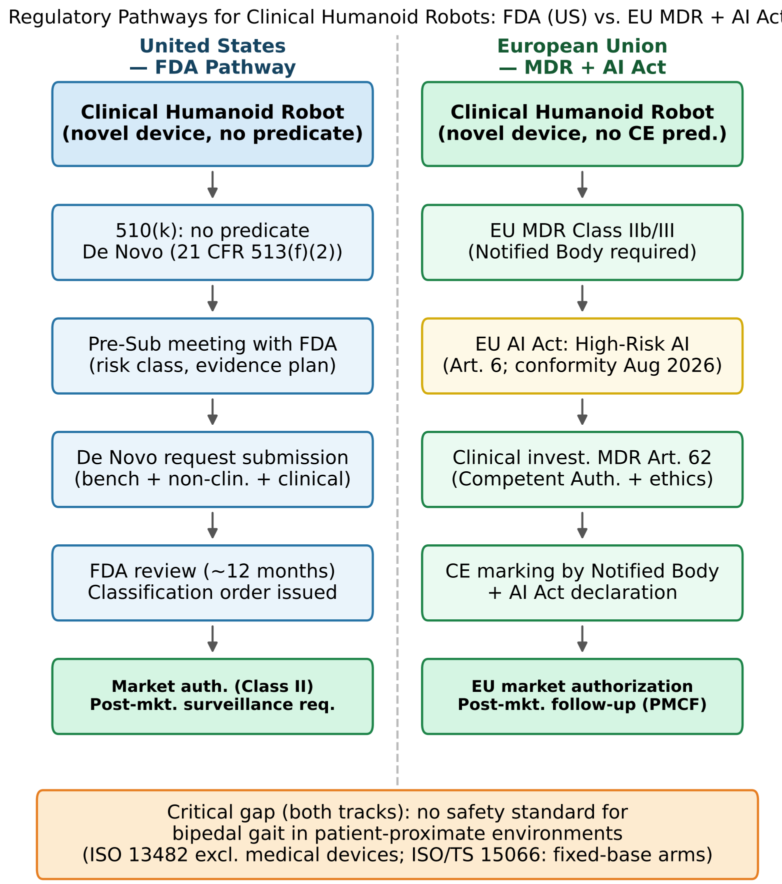

Fig. 5. Regulatory pathways for clinical humanoid robots: FDA De Novo (US, left) and EU MDR plus EU AI Act (EU, right). Both tracks require the same foundational evidence—safety standards for bipedal gait in patient-proximate environments—that does not yet exist (bottom banner). FDA De Novo is identified as the most plausible first-clearance pathway; EU MDR plus AI Act dual compliance represents the EU parallel track.

The path to the first regulatory clearance requires three sequential steps.

Step 1: Performance standards definition. A joint effort among humanoid manufacturers, academic medical centers (UCSD, JHU), and FDA's Center for Devices and Radiological Health (CDRH) is required to define the Special Controls that any De Novo submission must meet: fall prevention performance requirements, patient contact force limits for dynamic bipedal scenarios, teleoperation latency bounds, and AI/ML software validation methodology consistent with FDA's Good Machine Learning Practice principles and the December 2024 PCCP guidance.

Step 2: IRB feasibility study generating clinical evidence. A registered, IRB-approved feasibility study in a specific narrow indication generates the clinical evidence required for a De Novo submission. This study must be designed from the outset with regulatory submission in mind: pre-specified safety endpoints and a defined safety monitor protocol.

Step 3: De Novo submission establishing a regulatory predicate. The De Novo application proposes a new product code for bipedal humanoid clinical assistants with Special Controls from Steps 1 and 2. First clearance establishes a predicate device, enabling subsequent 510(k) applications.

For the EU, the parallel path requires: an ISO/TC 299 scope revision of ISO 13482 to include clinical humanoids; EU MDR Class IIa conformity assessment; and EU AI Act compliance under high-risk AI obligations (effective August 2026 for new AI systems; August 2027 for AI embedded in regulated medical products). Regulatory precedents for navigating layered framework compliance exist in the surgical robotics domain.

6.5 Ethical Considerations

Dignity of care. Physical care interactions require that the patient feels cared for, not processed. The SPRING project finding that participants cited eye-level conversational positioning as important for dignified interaction [2] provides direct empirical evidence that form factor affects perceived dignity of care. The uncanny valley phenomenon—the negative affective response elicited by human-like but not fully human entities [39]—is particularly relevant for humanoid design in care settings, where distress rather than curiosity is the cost of a poor perceptual response. Care robotics designers must incorporate patient dignity explicitly as a design criterion alongside technical performance metrics.

Liability and informed consent. No current healthcare liability framework assigns accountability for harm caused by an autonomous or semi-autonomous robotic action in a care setting. The ethical dimensions of robot care for elderly individuals—including dignity, autonomy, and emotional manipulation risks—require explicit frameworks before deployment [56]. Before any human-subject deployment, consent processes for humanoid-assisted care must be defined. These require deliberate engagement between legal, ethical, and clinical stakeholders before the first human-subject study is launched.

Data privacy. A humanoid operating in a patient room integrates cameras, microphones, depth sensors, physiological monitors, and, in future deployments, integration with EHR systems, into a single mobile platform in one of the most privacy-sensitive environments imaginable. No data governance framework specific to clinical humanoid sensor data currently exists.

Workforce framing. The evidence does not support a displacement narrative. No clinical role exists that a humanoid currently performs reliably enough to replace a trained nurse or clinician. The evidence-supported framing is task offloading: routine, physically demanding, or cognitively repetitive tasks that do not require clinical judgment could be offloaded, freeing clinicians for judgment-requiring activities. The administrator survey of Imtiaz and Khan [22] shows that the perceived risk of reduced human contact is a primary institutional barrier; communicating the task-offloading framing accurately to institutional decision-makers is as important as resolving the technical barriers.

7 Conclusion

As of March 2026, no bipedal humanoid robot holds FDA clearance or CE marking for any clinical application. No registered Phase I, II, or III clinical trial has enrolled patients for a study involving a full-size bipedal humanoid platform. The systematic search underlying this review identified 14 qualifying papers across 2022–2026—a literature sparse enough to read in its entirety, concentrated on a single commercial platform (Unitree G1) and, for the clinically most advanced domain, on a single research cluster (UCSD Yip Laboratory and Johns Hopkins). That concentration is not evidence of a field in failure; it is evidence of a field in its earliest phase of organized research. The infrastructure that would enable distributed clinical investigation—accepted safety standards for bipedal patient-contact environments, IRB-validated study protocols, and regulatory pathways with established Special Controls—does not yet exist, and its absence explains the concentration far better than any deficit of platform capability or clinical motivation.

The contribution of this review is to document that gap with specificity sufficient to act on it. Prior reviews of healthcare robotics, including the closest comparative work in the literature [17], either predate the LLM/VLA inflection that is the enabling technical development of the current period, or treat humanoid and non-humanoid platforms interchangeably, obscuring the platform-specific barriers and advantages that define the clinical deployment problem. The synthesis here is the first to integrate the full 2022–2026 humanoid-specific literature with a commercial and academic platform capability assessment, a regulatory pathway analysis, and a per-domain TRL readiness evaluation grounded in clinical deployment criteria.

The field is not uniform in its readiness. Three roles have a credible path to first human-subject validation—TRL 5—within five years: a telemedicine and telepresence clinical assistant operating in fully teleoperated mode, which minimizes autonomous patient-contact risk and maps directly onto the established regulatory logic of physician-controlled surgical robotics; a rehabilitation movement demonstrator that exploits the mirror neuron mechanism established by Nguyen et al. [41] and the clinical acceptability demonstrated by Sobrepera et al. [58] without requiring any patient contact, rendering the bipedal gait safety gap irrelevant to this role; and a structured eldercare companion and physical assistant following the hybrid autonomous/teleoperated architecture demonstrated by RHP Friends [6] and the deployment methodology established by the SPRING project [2]. These are not speculative roles. They are identified by triangulating the strongest evidence in the literature, the most tractable technical barriers, and the lowest regulatory risk profile available to each application.

Two gaps are more consequential than the others identified in Section 5. The first is the absence of any accepted performance standard for bipedal humanoid robots in patient-proximate environments—specifically for the four categories of gait-induced contact events that ISO/TS 15066 was not designed to address. This gap does not require a breakthrough to close; it requires a biomechanical characterization study and the standards development process. But no IRB will approve physical patient-contact studies without it, and no regulatory agency can specify Special Controls without it. It is the single most globally blocking gap in the field. The second is the absence of registered human-subject feasibility studies. Cho et al. [16] advance the state of the art by completing a cadaveric surgical study under a defined IRB protocol—the step immediately preceding first-in-human research. UC San Diego and Johns Hopkins, which together hold all three papers in the clinical procedures domain, are the natural sites for this next step. There is no institutional reason, beyond the gait safety gap, why this study cannot begin within the current funding cycle.

The question confronting the field is not whether humanoid robots can play a useful role in healthcare. The evidence in this review establishes that they can—not at the level of clinical deployment, but at the level of demonstrated laboratory capability in procedures and care tasks that have genuine clinical value. The question is whether the research community, device manufacturers, clinical institutions, and regulatory agencies will commit to the specific, sequential, methodical process required to convert that laboratory evidence into clinical evidence, and clinical evidence into regulatory precedent. That process—safety characterization, IRB feasibility study, De Novo submission, and predicate device establishment—is well understood in medical device development. Applying it to a genuinely novel platform class, with novel form-factor dynamics and novel AI system integration, requires institutional investment and coordination that has not yet materialized at scale. The evidence base is ready for the next step. The infrastructure to take it is the work of the next five years.

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